

UNIVERSITEIT VAN PRETORIA / UNIVERSITY OF PRETORIA
FAKULTEIT NATUURWETENSKAPPE / FACULTY OF SCIENCE

DEPARTEMENT WISKUNDE EN TOEGEPASTE WISKUNDE
DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Calculus WTW 158
Eksamen / Examination

Interne Eksaminator/ Internal Examiner: Prof JC Engelbrecht

Eksterne Eksaminator/External Examiner: Dr AF Harding

November 2004

Tyd / Time: 45 min

Punte / Marks: 30

VAN / SURNAME: _____

VOORNAME / FIRST NAMES: _____

STUDENTENOMMER / STUDENT NUMBER: _____

HANDTEKENING / SIGNATURE: _____

**PUNTE
MARKS**

LEES DIE VOLGENDE INSTRUKSIES READ THE FOLLOWING INSTRUCTIONS

- | | |
|---|--|
| 1. Die vraestel bestaan uit bladsye 1 tot 5. Kontroleer of jou vraestel volledig is. | 1. The paper consists of pages 1 to 5. Check whether your paper is complete. |
| 2. Doen alle krapwerk op die leë bladsye. Dit word nie nagesien nie. | 2. Do all scribbling on the empty pages. It will not be marked. |
| 3. Toon alle bewerkings. | 3. Show all working details. |
| 4. As jy meer as die beskikbare ruimte vir 'n antwoord nodig het, gebruik dan die leë bladsye en trek 'n raam daarom . | 4. If you need more than the available space for an answer, use the empty pages and please indicate it clearly by framing it. |
| 5. Geen potlood of Tippex nie. | 5. No pencil or Tippex. |

1. As $f''(x) = \frac{1}{x^3}$, $f(1) = 2$ en $f'(1) = 1$, bepaal $f(x)$.
 If $f''(x) = \frac{1}{x^3}$, $f(1) = 2$ and $f'(1) = 1$, find $f(x)$.

[3]

2. Neem aan dat f en g differensieerbare funksies is, waarvan ons slegs die inligting in die tabel beskikbaar het: / Assume that f and g are differentiable functions of which we only know the data in the following table:

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
0	4	2	3	0

- 2.1 Laat $h(x) = \frac{f(x)}{g(x)}$. Bereken $h'(0)$.

Let $h(x) = \frac{f(x)}{g(x)}$. Compute $h'(0)$.

[2]

- 2.2 Laat $k(x) = \sqrt{f(x) + x^2}$. Bereken $k'(0)$.

Let $k(x) = \sqrt{f(x) + x^2}$. Compute $k'(0)$.

[2]

3. Laat / Let $f(x) = \frac{4-x}{4+x}$.

- 3.1 Bepaal die horisontale en vertikale asimptote van f (indien enige). Toon alle berekeninge.

Determine the horizontal and vertical asymptotes of f (if any). Show all computations.

[3]

3.2 Gee die interval(le) waarop f daal. Toon alle berekeninge.

Give the interval(s) where f is decreasing. Show all computations.

[2]

4. Bewys dat $\frac{d}{dx} (\sinh x) = \cosh x$.

Prove that $\frac{d}{dx} (\sinh x) = \cosh x$.

[2]

5. As $y = \ln(x + 2y)$, bepaal $\frac{dy}{dx}$ in die punt $(1, 0)$.
If $y = \ln(x + 2y)$, determine $\frac{dy}{dx}$ in the point $(1, 0)$.

6.1 Bepaal / Determine [2]

$$\int \frac{x}{\sqrt{1-9x^2}} dx$$

6.2. Bepaal / Determine [2]

$$\int \sin^4 x \cos^3 x \, dx$$

[3]

7. Gegee die volgende Cartesiese vergelykings van twee reguit lyne:
Given the Cartesian equations of two lines:

$$L_1 : x = -6t, y = 1 + 9t, z = -3t$$

$$L_2 : x = 1 + 2s, y = 4 - 3s, z = s$$

- 7.1 Toon aan dat L_1 en L_2 ewewydig is.
Show that L_1 and L_2 are parallel.

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- 7.2 Bepaal 'n Cartesiese vergelyking vir die vlak waarin L_1 en L_2 lê.
Find a Cartesian equation for the plane containing L_1 and L_2 .

[2]

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8. Gegee drie punte $A(2, 3, -1)$, $B(0, -4, 2)$ en $C(2, 2, -3)$.
Given three points $A(2, 3, -1)$, $B(0, -4, 2)$ and $C(2, 2, -3)$.

- 8.1 Bepaal die parametriesse vergelykings van die reguit lyn deur A en B. / *Find the parametric equations of the straight line through A and B.*

[3]

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- 8.2 Bepaal die area (oppervlakte) van die driehoek ABC . / *Find the area of the triangle ABC .*

[2]

[2]